

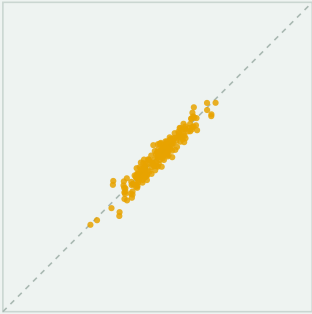
When two definitions share people, their noise cancels block by block

LDSC's 200-block jackknife, paired block by block across two definitions of one condition

● one genome block, both definitions — one block's difference ■ blocks paired at random

a Nested definitions

depression: sleep apnoea vs any sleep



across: hospital records
up: any sleep disorder

blocks agree, $r = 0.96$

paired

SE 0.007

b Partly shared cases

PTSD: constipation vs all of K59



across: K59.0 or laxatives
up: all of K59

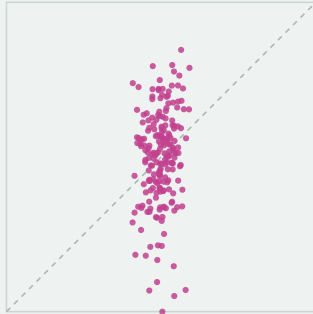
blocks agree, $r = 0.39$ 3 off scale

paired

SE 0.055

c Almost nothing shared

PTSD: EHR vs clinical depression

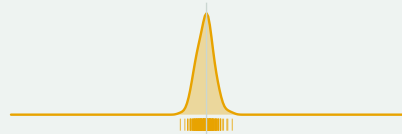


across: depression, EHR
up: depression, clinical

blocks agree, $r = 0.16$ 1 off scale

paired

SE 0.052



pairing broken

SE 0.035



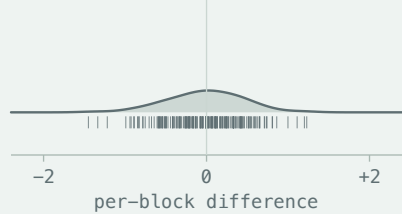
pairing broken

SE 0.070

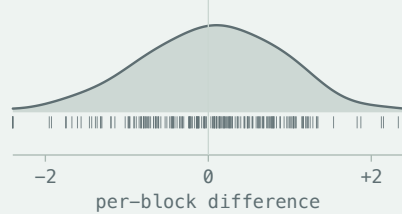


pairing broken

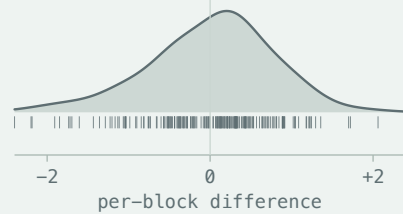
SE 0.055



per-block difference



per-block difference



per-block difference

READING THIS

Nested definitions: pairing the blocks cuts the standard error of the difference from 0.035 to 0.007.

Partly shared cases: from 0.070 to 0.055. Almost nothing shared: from 0.055 to 0.052.

Shared people, not sample size, decide how small a definition effect can be seen.